

ABB Manufacturing Engineering Drawing Standard

Manufacturing Tools, Jigs, Fixtures, Gauges, and Prototype Components

1. Drawing Philosophy

The engineering drawing is the official manufacturing document.

The CAD model shall represent the intended final geometry and shall be used as the source for drawing creation. The released PDF drawing shall fully define the part for personnel who do not have access to the CAD model.

If the part is not manufactured to the drawing, the part is not correct.

Drawings shall be:

- Fully dimensioned
- Clear and legible
- Practical for manufacturing
- Practical for inspection
- Suitable for First Article Inspection (FAI)

The drawing should communicate design intent without unnecessary complexity.

2. Drawing Format

All released drawings shall be provided as PDF files.

CAD files may be used internally for manufacturing, programming, and revision control, but the PDF drawing is the official manufacturing document.

3. Units

All drawings shall use:

INCHES

Metric dimensions shall not be used unless specifically required.

4. Title Block Requirements

The title block shall include:

- Part Title
- Material
- Drawing Size
- Revision
- Scale
- Sheet Number
- Part Number

Approval information:

- Drawn By
- Engineering Approval
- Manufacturing Engineering Approval
- Approval Initials
- Approval Date

The title block shall also include:

- General tolerances
 - Drawing revision information
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5. Drawing Sheets

Complex parts should use multiple sheets when required for clarity.

Typical format:

Sheet 1 — Machining Details

Contains:

- Machining dimensions
- Feature locations
- Tolerances
- Section views
- Manufacturing notes

Sheet 2 — Laser Engraving Details

Contains:

- Part number engraving location
- Revision marking
- Identification requirements

Most manufactured tooling components should include part identification engraving when practical.

6. Dimensioning Philosophy

All manufactured features shall be fully defined.

Dimensions shall be arranged to maintain drawing clarity.

Avoid excessive leaders, overlapping dimensions, and clutter.

Preferred practices:

- Break extension lines when necessary
- Use ordinate dimensions for complex features
- Use detail views when appropriate
- Maintain logical dimension flow

The drawing should be easy to manufacture and easy to inspect.

7. Datum Philosophy

Functional datums shall be used when required.

For most tooling components, the primary reference is typically an edge or surface of the part.

A datum symbol is not required unless GD&T functionality requires it.

The preferred method is to select a practical reference edge and dimension features from that location.

8. General Tolerances

Default drawing tolerance:

.000

Dimensions shall be shown to the required precision.

Typical expectations:

- General machining dimensions: ± 0.005
- Precision fits: .0000 dimensions when required

Four decimal place dimensions shall be reserved for precision applications.

9. Hole Callouts

All holes shall use diameter symbols.

Fractional hole callouts shall not be used.

Preferred:

Ø0.250 THRU

Not:

1/4 THRU

10. Hole Quantity Callouts

Use PLACES for repeated features.

For three or fewer identical features:

Example:

3 PLACES Ø0.250 THRU

For more than three identical features:

Example:

Ø0.250 THRU TYP

Do not use X notation.

Incorrect:

4X Ø0.250 THRU

11. Hole Location

Coordinate dimensioning is the preferred method for locating holes.

Hole locations should typically be referenced from the selected part edge.

GD&T position callouts should only be used when required by engineering function.

12. Hole Identification

Every hole shall include:

- Center mark
- Appropriate centerline representation

Centerlines should be used where they improve understanding, especially for:

- Symmetric parts
- Pattern locations
- Rotational features

Do not dimension to imaginary centerlines unless the centerline is functionally important, as these features can be difficult to inspect.

13. Counterbores and Countersinks

Counterbores shall use standard symbols.

Do not use abbreviated terminology.

Example:

Ø0.250 THRU

Counterbore symbol: Ø0.500 X 0.250 DEEP

14. Chamfers

Standard chamfer callout:

CHAMFER 0.050 X 45°

Hole chamfers:

CHAMFER ALL HOLES UNIFORMLY

15. Edge Break Requirements

Unless otherwise specified:

BREAK ALL SHARP EDGES AND REMOVE ALL BURRS

All manufactured components shall be free of:

- Sharp edges
 - Burrs
 - Machining debris
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16. Section Views

Section views shall be used when necessary to clearly communicate internal features.

Section views shall follow the naming convention:

SECTION X-X

SECTION A-A

SECTION B-B

Section lines and labels shall clearly identify the referenced view.

17. GD&T Philosophy

GD&T shall only be applied when it provides meaningful engineering control.

The preferred drawing approach is:

- Coordinate dimensions
- Functional dimensions
- Practical inspection methods

Avoid unnecessary GD&T.

The following controls are generally not used unless specifically required:

- Profile tolerances
- Complex datum schemes
- Advanced GD&T controls

18. Form and Orientation Controls

When required by engineering function:

Flatness: 0.002

Parallelism: 0.002

Perpendicularity: 0.002

These controls should only be added when function requires them.

19. Surface Finish

Surface finish requirements shall only be specified when required by function.

Do not add unnecessary finish requirements.

20. Manufacturing Design Philosophy

Drawings shall support Design for Manufacturing (DFM).

Designers should:

- Use common tooling
- Use standard cutters

- Use standard thread sizes
- Avoid unnecessary small features
- Avoid fragile features
- Minimize machining setups

The goal is reliable manufacturing with reasonable cost.

21. First Article Inspection

The drawing shall support First Article Inspection.

Inspectors should be able to measure each drawing dimension and record the actual measured value next to the nominal requirement.

Dimensions should be arranged to support inspection workflow.

22. Drawing Release Checklist

Before release verify:

- ☐ CAD model matches drawing
 - ☐ PDF drawing generated from current CAD revision
 - ☐ All features fully defined
 - ☐ Dimensions are readable and organized
 - ☐ Hole locations are clear
 - ☐ All holes contain center marks
 - ☐ Material specified
 - ☐ Part number specified
 - ☐ Revision identified
 - ☐ Laser engraving requirements included when applicable
 - ☐ Special instructions included
 - ☐ Manufacturing Engineering approval complete
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Standard Statement

A good manufacturing drawing is not the drawing with the most GD&T.

A good manufacturing drawing is one that allows a machinist to make the correct part and allows an inspector to verify it without interpretation.

The drawing shall prioritize:

Function → Manufacturing → Inspection → Documentation